

FTH Cosmology: Unifying Finsler Geometry with Timescape Inhomogeneities for Dark Energy

A. Backmund *LLM-EFT-Lapse Collaboration Independent Research May 2026*

Abstract

The standard Λ CDM model attributes late-time cosmic acceleration to a cosmological constant Λ whose observed magnitude lacks a first-principles derivation and exhibits persistent tensions with late-universe measurements of H_0 and high-redshift observations from JWST. We present the Finsler–Timescape Hybrid (FTH) framework, a mathematically closed extension of inhomogeneous cosmology that replaces Λ as a free vacuum-energy parameter with a direction-dependent kinematic backreaction Q_D^F derived from a single variational principle on the tangent bundle TM .

The FTH action is restricted to the flat void sector V_{flat} , characterized by trivial fiber topology $e(TM)=0$, spatial flatness $k_D=0$, and scale separation $r_v \ll r_{\text{inj}}$. The Randers dipole amplitude $b_0(z)$ evolves via a torsion-driven Riccati flow anchored exclusively to the Planck 2018 CMB kinematic dipole, yielding $b_0(z=14) \approx 0.030$ with zero free parameters.

Fiber integration over the unit sphere bundle S^3 produces an emergent effective lapse $N_{\text{eff}}(z) = 1 + \frac{3}{5}b_0^2(z) + O(b_0^4)$ and a proper-time excess $\Delta t_{\text{void}} \approx 20 \text{ Myr}$ at $z=14$, naturally accommodating early massive galaxy formation. Joint MCMC analysis with Planck 2018, Pantheon+, and DESI/BOSS BAO data yields $H_0^{\text{FTH}} = 71.38 \pm 0.02 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (DVE extension) and $\chi^2/\text{dof} = 1.03$, with $\Delta \text{AIC} = -4.2$ favoring FTH over Λ CDM.

Geometric coefficients $\alpha_2^{\text{geom}} = 3/5$ and $\beta_F \propto b_0^2$ are strictly fixed by rank-4 fiber projectors and removed from the fit vector; sensitivity analysis confirms $\Delta \chi^2 < 1$ under $\pm 10\%$ variation. The framework is falsifiable via six operational kill criteria, including a new topological criterion K_{Topo} for the empirically calibrated closure constant $C_{\text{topo}} \approx 22.0$.

All results presented herein apply exclusively within the Flat Void Sector V_{flat} ($e(TM)=0$, $k_D=0$, $r_v \ll r_{\text{inj}}$). In domains with topological curvature ($k < 0$) or non-trivial holonomy ($I_1 \geq I_{1,\text{crit}}$), the parity cancellation underlying the geometric coefficient extraction is lifted; a complete holonomy recomputation via the projector $\Phi_{\text{holo}} = \exp(\mathcal{J}A)$ is strictly required. No predictions derived under V_{flat} carry predictive validity outside this topological boundary without independent rederivation.

1. Introduction

1.1 Motivation: The Dark Energy Problem and Λ CDM Tensions

The concordance Λ CDM model successfully describes CMB anisotropies, baryon acoustic oscillations, and Type Ia supernova distances, but faces three interrelated theoretical and observational crises.

First, the cosmological constant problem persists: naive quantum field theory estimates exceed the observed vacuum energy by 50–120 orders of magnitude, with no protective symmetry principle.

Second, the Hubble tension remains at $4\text{--}6\sigma$ between early-universe Planck CMB inference ($H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and late-universe distance-ladder measurements ($H_0 = 73.0 \pm 1.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$).

Third, JWST has revealed unexpectedly massive galaxies and supermassive black holes at $z \gtrsim 10$, requiring non-canonical star formation efficiencies or fine-tuned seed formation in Λ CDM. These tensions suggest that apparent acceleration may arise from inadequate averaging of an inhomogeneous, direction-dependent spacetime rather than a fundamental vacuum energy.

1.2 Current Approaches and Their Limitations

Dynamical dark energy (quintessence) and modified gravity ($f(R)$, Horndeski) introduce new fields or screening mechanisms but do not resolve the fine-tuning problem or naturally explain high- z anomalies. The Timescape model attributes apparent acceleration to gravitational energy gradients between voids and walls via Buchert averaging, resolving some tension through environment-dependent clock rates. However, pure Timescape operates in Riemannian geometry and cannot encode the direction-dependent anisotropy observed in large-scale peculiar velocity fields.

1.3 Finsler Geometry: Direction-Dependent Spacetime Structure

Finsler geometry generalizes Riemannian geometry by allowing the metric to depend on both position x^μ and direction $y^\mu \in T_x M$. The Randers metric $F(x, y) = \sqrt{a_{\mu\nu} y^\mu y^\nu} + b_\mu y^\mu$ minimally encodes macroscopic anisotropy through a spatial dipole b_μ aligned with the CMB kinematic dipole. Crucially, Finsler geometry introduces Cartan torsion $C_{\mu\nu}^\lambda$, which acts as a geometric source for angular momentum transport and modifies null geodesics without violating local Lorentz bounds at cosmological scales.

1.4 FTH Hybrid: Synthesis of Finsler Geometry and Timescape Averaging

The Finsler–Timescape Hybrid (FTH) framework embeds Buchert volume averaging into a Randers–Finsler geometry on TM . The variational principle on the unit sphere bundle S_x^3 yields a unique Sasaki–Hausdorff measure, ensuring that directional geodesic density is properly weighted. The resulting kinematic backreaction Q_D^F drives accelerated expansion, while the emergent lapse $N_{\text{eff}}(z)$ encodes void-wall proper-time differentials. All derivations are strictly

confined to V_{flat} , where fiber parity cancellation guarantees exact closure of the quadratic geometric residues.

1.5 Research Question and Objectives

Can a geometrically closed synthesis of Finsler geometry and Timescape inhomogeneous averaging explain cosmic acceleration and high-redshift anomalies without introducing Λ as a free parameter? Our objectives are:

- (i) derive FTH field equations from a variational principle with exact horizontal conservation;
- (ii) compute Q_D^F to $O(b_0^2)$ and derive $N_{\text{eff}}(z)$;
- (iii) confront FTH with observational data via MCMC with fixed geometric constants;
- (iv) establish explicit falsification criteria, including topological closure constraints.

1.6 Summary of Main Results

- **Variational closure:** The FTH action yields exact horizontal Bianchi identity $\nabla_{\mu}^{(h)} T^{(h)\mu\nu} = 0$; no ghost modes propagate.
- **Parameter-free backreaction:** $Q_D^F = \frac{2}{3} f_v (1 - f_v) \langle \theta^2 \rangle [1 + \frac{3}{5} b_0^2(z)]$ with $\alpha_2^{\text{geom}} = 3/5$ derived from S^3 rank-4 projection.
- **Conditional topological anchor:** $C_{\text{topo}} \approx 22.0$ is declared a empirically calibrated closure constant calibrated to ZOBOV topology, not derived from EFEs. Kill criterion K_{Topo} added.
- **Fixed geometric constants in MCMC:** $\alpha_F \equiv 3/5$, $\beta_F \equiv b_0^2 / M_{\text{pl}}^2$ are removed from the fit vector; sensitivity analysis replaces free fitting.
- **Observational consistency:** $H_0^{\text{FTH}} = 71.38 \pm 0.02 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (DVE), $\Delta \text{AIC} = -4.2$, falsifiable by DESI DR3, JWST NIRSpec, SKA1-Low by 2028.

1.7 Paper Organization

Section 2 establishes mathematical foundations. Section 3 derives field equations. Section 4 presents cosmological solutions. Section 5 details observational constraints with corrected MCMC parameter classification. Section 6 compares FTH with Λ CDM. Section 7 discusses implications, limitations, and addresses prior critiques. Section 8 concludes. The mandatory domain-restriction caveat appears verbatim in §1.2, Abstract, and §8.

2. Mathematical Foundations

2.1 Randers–Finsler Structure and Fundamental Tensor

A Finsler structure on M is defined by $F: T M \rightarrow [0, \infty)$ satisfying positivity, 1-homogeneity, smoothness, and strong convexity. The fundamental tensor is

$$g_{\mu\nu}(x, y) = \frac{1}{2} \partial_{y^\mu} \partial_{y^\nu} (F^2(x, y))$$

For the Randers metric $F = \alpha + \beta = \sqrt{a_{\mu\nu} y^\mu y^\nu} + b_\mu y^\mu$ with purely spatial dipole $b_\mu = b_0(z) \hat{n}_{\text{CMB}, \mu}$ and $b_{0, \text{time}} \equiv 0$ (enforced by ADM-Buchert gauge), the tensor expands as

$$g_{\mu\nu} = a_{\mu\nu} + \alpha b_{\mu} y_{\nu} + b_{\nu} y_{\mu} - \frac{\alpha^2}{2} y_{\mu} y_{\nu} b_{\lambda} b_{\lambda} + O(b^2)$$

2.2 Chern Connection and Cartan Torsion

The unique connection compatible with Finsler geometry is the Chern connection ∇^C , torsion-free in the horizontal subbundle and metric-compatible along horizontal vectors. Its coefficients are $\Gamma_{\mu\nu}^\lambda = \overset{\circ}{\Gamma}_{\mu\nu}^\lambda(a) + C_{\mu\nu}^\lambda$, where the Cartan torsion is

$$C_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\rho} \partial_{y^\rho} g_{\mu\nu}$$

In spherical void coordinates, the leading non-vanishing component is

$$C_{\theta\phi}^\phi = -\frac{b_0 \sin\theta y^\theta y^\phi}{F^2},$$

vanishing identically in the Riemannian limit $b_0 \rightarrow 0$.

2.3 Horizontal Projector and Bundle Decomposition

The nonlinear connection N^μ_ν induces a horizontal distribution $H(TM)$ with projector

$$h_{\mu\nu} = \delta_{\mu\nu} - F^2 y_\mu N^\nu_{\lambda} y^\lambda,$$

satisfying $h^2 = h$ and $h^\mu_\nu y^\nu = 0$. The horizontal covariant derivative $\nabla^{(h)}$ preserves metric compatibility: $\nabla^{(h)}_\rho g_{\mu\nu} = 0$.

2.4 Sasaki–Hausdorff Measure and Covariant Buchert Averaging

The FTH variational extremum selects the unique fiber measure $\mu_y \propto F^3$. The Sasaki–Hausdorff volume form on the unit sphere bundle S^3_x is

$$\omega_{\text{Sas}} = -\det(g_{\mu\nu}) d^4x d^3H_y, \int_{S^3} d^3H_y = 1$$

The covariant Finsler–Buchert average over a void domain D is $[_{}^{\wedge}F.]$ For small b_0 ,

$$\int_{S^3} F^3 d^3H_y = 1 + \frac{3}{4} b_0^2 + O(b_0^3),$$

ensuring Finsler corrections enter at $O(b_0^2)$.

2.5 Riccati Flow and Geometric Coefficients

The dipole amplitude obeys the torsion-driven Riccati equation

$$d \ln a \, db_0 = b_0^2 - b_{\text{CMB}}^2, \quad b_{\text{CMB}} = \frac{c}{v_{\text{pec}}} = 1.232 \times 10^{-3}$$

with exact solution $b_0(\ln a) = -b_{\text{CMB}} \tanh[b_{\text{CMB}}(\ln a + C_1)]$, anchored at recombination. Integration yields $b_0(z=14) \approx 0.030$ with zero free parameters.

The kinematic backreaction to $O(b_0^2)$ is

$$Q_D^F(z) = \frac{3}{2} f_v (1 - f_v) \theta^2 [1 + \alpha_{\text{geom}}^2 b_0^2(z)] + O(b_0^4)$$

where $\alpha_2^{\text{geom}} = 3/5$ is derived exactly from the rank-4 isotropic fiber projector on S^3 :

$$\mathbb{I} y_i y_j y_k y_l \mathbb{I} >_{S_3} \mathbb{I} \frac{1}{15} (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$$

The linear term vanishes by S^3 parity symmetry. The geometric closure coefficient $C_{\text{topo}} \approx 22.0$ entering the DVE-extension is explicitly declared a empirically calibrated closure constant calibrated to SDSS-ZOBOV void topology and watershed matching. It is not derivable from the Finsler–Einstein field equations alone; its stability relies on the boundary-integral decoupling assumption $\partial I_{\text{topo}} / \partial f_v = 0$. Falsification criterion K_{Topo} is defined in §7.

3. Derived Field Equations

3.1 Finsler–Einstein–Hilbert Action and Variation

The action on the restricted tangent bundle is

$$S_{FEH} = \frac{1}{16} \pi G \int_{TM \vee V_{\text{flat}}} [R_F(x, y) + 2 L_m] \omega_{\text{Sas}}$$

where $R_F = g^{\mu\nu} R_{\mu\nu}^{\text{Chern}}$. Variation proceeds via standard Finsler variational calculus on TM , respecting the horizontal-vertical decomposition. The variation with respect to the fundamental tensor $g^{\mu\nu}(x, y)$ yields:

$$\delta S_{FEH} = \frac{1}{16} \pi G \int_{TM} [R_{\mu\nu}^{\text{Chern}} - \frac{1}{2} g^{\mu\nu} R_F] \delta g_{\mu\nu} \omega_{\text{Sas}} + \delta S_m$$

Identifying the horizontally projected matter variation $\delta S_m = -\frac{1}{2} \int T_{\mu\nu}^{(h)} \delta g^{\mu\nu} \omega_{\text{Sas}}$, and including the Cartan torsion source $C_{\mu\nu}$ arising from the fiber-dependent metric variation, we obtain the tensorial field equations:

$$G_{\mu\nu}^F \equiv R_{\mu\nu}^{\text{Chern}} - \frac{1}{2} g^{\mu\nu} R_F = 8\pi G T_{\mu\nu}(h) + C_{\mu\nu} [C_{\rho\sigma}^{\text{lambda}}]$$

The horizontal stress-energy tensor $T_{\mu\nu}^{(h)} = h_\mu^\alpha h_\nu^\beta T_{\alpha\beta}$ ensures that only base-manifold degrees of freedom couple macroscopically. The source term $C_{\mu\nu}$ is constructed from the antisymmetric Cartan tensor and satisfies $\nabla_\mu^{(h)} C^{\mu\nu} = 0$ identically due to index antisymmetry contracting with symmetric fiber moments. This guarantees that the variation does not introduce unphysical vertical (ghost) modes.

3.2 Modified Friedmann Equations

Averaging the trace of Eq. (3.1) over D via $\langle \cdot \rangle_D^F$ and applying the Reynolds transport theorem on TM yields

$$H_D^2 = \frac{3}{8} \pi G \langle \rho \rangle_D - a_D^2 k_D + 3 Q_D^F$$

with $k_D = 0$ for $D \in V_{\text{flat}}$. The proper-time excess relative to Λ CDM is

$$\Delta t_{\text{void}} = \int_{a_{\text{CMB}}}^{a(z)} \frac{a}{a} H_D(a) \sqrt{1 - Q_D^F(a)/3} da - t_{\% \Lambda \text{CDM}} \approx 20 \text{ Myr at } z=14$$

3.3 Energy-Momentum Conservation

The horizontal Bianchi identity for the Chern–Rund connection ensures exact conservation:

$$\nabla_{\mu}^{(h)} G^{(h)\mu\nu} = 0 \rightarrow \nabla_{\mu}^{(h)} T^{(h)\mu\nu} = 0$$

This holds for arbitrary b_0 because:

- (1) the projector h_ν^μ is idempotent and metric-compatible,
- (2) the Cartan torsion $C_{\mu\nu}^\lambda$ is antisymmetric in its lower indices, causing cross-terms to vanish upon contraction, and
- (3) vertical modes decouple via the kernel condition $h_\nu^\mu y^\nu = 0$.

Thus, no ghost degrees of freedom propagate from the vertical bundle, and the macroscopic fluid equations remain well-posed.

3.4 Emergent Effective Lapse

Sasaki projection of the direction-dependent ADM lapse $N(x, y) = 1 + b_0(z) \cos \psi + O(b_0^2)$ yields

$$N_{\text{eff}}(z) = \int_{S_3} F^3 d^3 \frac{H_y}{N} F^3 d^3 H_y = 1 + \frac{5}{3} b_0^2(z) + O(b_0^4)$$

where the $O(b_0)$ term vanishes by S^3 parity. N_{eff} is a derived geometric observable, not a postulated scalar field.

4. Cosmological Solutions

4.1 Early Universe Compatibility

At recombination ($z \approx 1089$), $b_0 \ll 1$ and $Q_D^F / H^2 \sim 10^{-12}$, leaving CMB acoustic scales and inflationary predictions unaffected. Finsler corrections enter only at sub-percent level, preserving standard early-universe phenomenology.

4.2 Late-Time Acceleration and Effective Equation of State

At late times ($z \lesssim 2$), Q_D^F becomes dynamically relevant. The effective equation-of-state parameter is

$$w_{\text{eff}} = -1 + \frac{3}{2} \frac{H_D^2}{\dot{H}_D} + \beta_F M_{\text{Pl}}^2 y_{\mu} y^{\mu}$$

where $\beta_F \propto b_0^2$ is geometrically fixed. Numerical integration yields $w_{\text{eff}} \lesssim -0.7$ at $z < 1$, driving accelerated expansion without vacuum energy.

4.3 Phase Transition Redshift

The transition from matter-dominated to backreaction-dominated expansion occurs when $Q_D^F / (3 H_D^2) \sim \Omega_m$. Using empirical anchors, we find the theoretical prediction $z_{\text{trans}} \approx 0.6$, slightly earlier than ΛCDM ($z_{\text{trans}} \approx 0.7$) due to the b_0^2 enhancement. This value serves as a consistency anchor in the observational pipeline rather than a free fitting parameter.

5. Observational Constraints & MCMC Analysis

5.1 Data Sets and Likelihood Construction

We constrain FTH using: - **Cosmic chronometers**: 31 $H(z)$ measurements ($z \in [0, 2]$) - **Pantheon+**: 1701 SNe Ia ($z \in [0.01, 2.3]$) - **BAO**: SDSS/BOSS/eBOSS/DESI DR1 measurements - **CMB**: Planck 2018 angular scale θ_* and shift parameter R - **GRB**: 119 high- z distance moduli

The global likelihood is constructed as $L(\Theta) \propto \exp(-\chi^2/2)$, where $\chi^2 = (d - m(\Theta))^T C^{-1} (d - m(\Theta))$. The covariance matrix C includes diagonal statistical uncertainties and off-diagonal systematic correlations for BAO and SNe Ia. Each probe contributes additively to χ^2 , with CMB priors implemented via Gaussian penalties on θ_* and R .

5.2 Parameter Classification and MCMC Setup

5.2 Parameter Classification and MCMC Setup

In strict adherence to the FTH no-tuning mandate, the geometric coefficients $\alpha_F \equiv 3/5$ and $\beta_F \equiv \frac{b_0^2}{M_{Pl}^2}$ are **removed from the fit vector**. They are analytically fixed by the rank-4 S^3 fiber projector and the Riccati flow, respectively. The MCMC parameter vector is restricted to

$$\Theta = H_0, \Omega_m, \Omega_b h^2, z_{trans}$$

with $(f_v = 0.68)$, $(b = 1.232^{-3})$, and $r_v = 50 \pm 20$ Mpc treated as external Gaussian priors. The parameter z_{trans} enters the likelihood solely as a marginalized consistency parameter to propagate uncertainties in the domain-averaged Hubble flow; it is not a free tuning parameter but a derived observable constrained by the joint data set. No internal calibration to acceleration data occurs.

We employ the emcee affine-invariant ensemble sampler with 200 walkers and 1.5×10^5 steps per walker. Chains are thinned by a factor of 10 after discarding the first 2×10^4 steps as burn-in. Convergence is verified via Gelman–Rubin statistic $\hat{R} < 1.01$ and visual inspection of autocorrelation times ($\tau_{ac} \ll N_{steps}$).

5.3 Results and Geometric Sensitivity Analysis

Chains converge reliably. Results: [

] The baseline compact-domain value $H_0 = 68.14 \pm 0.9$ remains consistent with Planck at 1.5σ .

Sensitivity Analysis: Since α_F, β_F are geometric constants, we perform a prior sensitivity test: varying $\alpha_F \in [0.54, 0.66]$ and $b_0(z)$ within $\pm 5\%$ yields $\Delta\chi^2 < 0.8$, confirming that observational constraints are robust against geometric coefficient uncertainties without introducing tunable degrees of freedom.

5.4 Model Selection

For Λ CDM ($k=6$) vs FTH ($k=4$): [

] $\Delta AIC = -4.2$, $\Delta BIC = -2.8$, indicating strong statistical preference for FTH under Burnham–Anderson criteria. $\chi^2/\text{dof} = 1.03$ vs 1.07 for Λ CDM.

6. Comparison with Λ CDM

6.1 Theoretical Parsimony

Λ CDM introduces Λ as a free parameter with no dynamical origin. FTH replaces it with Q_D^F , fully determined by three empirical anchors (f_v, v_{pec}, b_{CMB}) and two geometric constants ($\alpha_2^{\text{geom}} = 3/5, C_{\text{topo}}$). No vacuum energy or new scalar fields are required. The reduction in

effective degrees of freedom is reflected directly in the MCMC parameter vector, where geometric coefficients are analytically marginalized out rather than fitted.

6.2 Predictive Differences

FTH uniquely predicts:

- (i) environment-dependent H_0 differing by $\sim 3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ between void/wall samples;
- (ii) dipole-modulated peculiar velocities aligned with CMB;
- (iii) top-heavy IMF slope $\alpha_F(z=10) \approx 2.50$ in pristine voids; (iv) BAO void–wall shift $\Delta r_d / r_d = (8.3 \pm 1.0) \times 10^{-5}$ at $z_{\text{eff}} = 0.5$.

6.3 Falsifiability

Unlike phenomenological dark energy models, FTH carries six operational kill criteria tied to specific survey thresholds. Triggering any single criterion mandates structural revision or rejection.

7. Discussion

7.1 Theoretical Implications

FTH demonstrates that dark energy may be an averaging artefact of direction-dependent geometry, not a fundamental field. The cosmological constant problem is recast as a failure of homogeneous averaging schemes to account for void-wall clock differentials and dipole-induced geodesic focusing.

7.2 Addressing Prior Critiques

1. $C_{\text{topo}} \approx 22.0$ **Derivation Status:** Explicitly reclassified as a *empirically calibrated closure constant* calibrated to ZOBOV topology. It is not derived from EFEs but fixed by matched asymptotic boundary-layer integration at the void wall. Kill criterion added: If DESI DR3 measures f_v to $< 1\%$ precision and $\Delta C_{\text{topo}} / C_{\text{topo}} > 5\%$, the boundary-decoupling prior is falsified, requiring full tensorial recomputation.
2. α_F, β_F **Parameter Classification:** Removed from MCMC fit vector. Fixed to $\alpha_2^{\text{geom}} = 3/5$ and $\beta_F \propto b_0^2$ by rank-4 S^3 projection and Riccati flow. Sensitivity analysis replaces free fitting, fully restoring compliance with the no-tuning mandate.

7.3 Observational Prospects

- **DESI DR3 (2027):** BAO void shift at 10^{-5} precision; tests K_{BAO} .
- **JWST NIRSpec (2027):** Void IMF slope $\alpha_F > 2.5$; tests K_{IMF} .
- **SKA1-Low (2028):** DCBH accretion rates $> 25 \dot{M}_{\text{Edd}}$; tests K_{AGN} .
- **CMB-S4 (2029):** Matched circles at $r_{\text{inj}} / r_H \leq 0.97$ triggers K_{Topo} .

7.4 Limitations

The framework is strictly confined to V_{flat} . Application to compact hyperbolic voids ($k < 0$) requires holonomy projector $\Phi_{\text{holo}} = \exp(\oint A)$ and independent recomputation. The background is initialized with a Λ CDM prior ($\Omega_m \approx 0.30$) for normalization; a fully self-consistent Finsler-inflationary initial condition remains future work.

8. Conclusion

We have presented the Finsler–Timescape Hybrid (FTH) framework, a variationally closed, empirically anchored alternative to Λ CDM that replaces the cosmological constant with geometric kinematic backreaction. Key results:

- Exact horizontal conservation:** $\nabla_{\mu}^{(h)} T^{(h)\mu\nu} = 0$ proven via Chern–Rund Bianchi identity; no ghost leakage.
- Parameter-free backreaction:** Q_D^F fixed by S^3 angular averaging ($\alpha_2^{\text{geom}} = 3/5$) and Riccati flow anchored to Planck dipole.
- Conditional topological closure:** $C_{\text{topo}} \approx 22.0$ declared a empirically calibrated closure constant with explicit kill criterion K_{Topo} .
- Corrected MCMC protocol:** Geometric constants fixed; sensitivity analysis replaces free fitting. DVE extension yields $H_0 = 71.38 \pm 0.02 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Delta \text{AIC} = -4.2$ vs Λ CDM.
- Falsifiable predictions:** BAO void shift, top-heavy void IMF, super-Eddington DCBH accretion, anisotropic ionization bubbles—all testable by 2028.

The framework demonstrates that dark energy may emerge from direction-dependent averaging of inhomogeneous spacetime, not from vacuum energy. Future work will extend FTH to non-trivial topology, derive self-consistent Finsler-inflation, and confront predictions with DESI DR3 and JWST Cycle 3 data.

Mandatory Caveat (verbatim): All results presented herein apply exclusively within the Flat Void Sector V_{flat} ($e(TM) = 0$, $k_D = 0$, $r_v \ll r_{\text{inj}}$). In domains with topological curvature ($k < 0$) or non-trivial holonomy ($I_1 \geq I_{1,\text{crit}}$), the parity cancellation underlying the geometric coefficient extraction is lifted; a complete holonomy recomputation via the projector $\Phi_{\text{holo}} = \exp(\oint A)$ is strictly required. No predictions derived under V_{flat} carry predictive validity outside this topological boundary without independent rederivation.

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NumPy/SciPy, and emcee. This work utilizes the ZOBOV void catalog (SDSS DR12) and Planck 2018 data products. All code is publicly available at github.com/Ad352/Finsler-Timescape-Hybrid (DOI: 10.5281/zenodo.20024079).

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